

Find My Polynomial

Fall 2024 ARML Power Contest

The Background

Alice and Bob are playing a game. Alice secretly chooses a polynomial $p(x)$, perhaps with some restrictions such as requiring all coefficients to be positive or requiring all roots to lie in a certain interval. Bob must try to figure out what polynomial Alice has chosen. Bob is allowed to ask Alice to evaluate her polynomial for any real input x , and Alice must answer with the correct value of $p(x)$. Such an exchange will be called a *question*. Depending on the restrictions on Alice's polynomial, Bob might have to ask many questions before he can find the polynomial, or he might be able to do so with just one question. Your challenge is to find out how many questions will be needed in each case.

For the purposes of this problem, a *polynomial* is an expression of the form $p(x) = a_k x^k + a_{k-1} x^{k-1} + \cdots + a_1 x + a_0$ where all of the coefficients a_i are real numbers and all exponents are nonnegative integers. This polynomial has *degree* k if and only if $a_k \neq 0$, otherwise the text will specify that the degree is *at most* k . Should it be necessary, the polynomial all of whose coefficients are zero will be said to have undefined degree. A *monomial* is a polynomial with just one non-zero term, $a_k x^k$, with $a_k \neq 0$. Alice and Bob both have access to a very fast computer that can solve any equation or system of equations to any desired degree of accuracy instantaneously.

You may use the fact that a polynomial of degree at most k that has $k + 1$ or more distinct roots must be the zero polynomial, $p(x) = 0$.

The Problems

There are 40 points available on this contest. The numbers at the end of each problem are the number of points that problem is worth. These are also indicated on the cover sheet, which should be the front page of your solution packet. Please make sure to write on only one side of the answer sheets.

1. Given the restriction that Alice must choose a monomial, $a_k x^k$, Bob can determine Alice's polynomial by asking just two questions. Bob asks Alice to compute $A = p(1)$ and $B = p(2)$. Given the values of A and B , determine Alice's polynomial. That is, compute the exact values of k and a_k in terms of A and B [2 pts.]
2. Show that even if it is known that Alice has chosen a monomial, it is impossible for Bob to determine it by asking just one question. That is, for any x that Bob might choose, and given that $p(x) = A$, find two different monomials that could be Alice's polynomial. [2 pts.]
3. Alice is restricted to choosing a polynomial of degree at most 1. Given the values $A = p(0)$ and $B = p(1)$, determine Alice's polynomial (in terms of A and B). [2 pts.]
4. Alice is restricted to choosing a polynomial of degree at most 2. Given the values $A = p(0)$, $B = p(1)$, and $C = p(-1)$, determine Alice's polynomial (in terms of A , B , and C). [2 pts.]

Given any $k + 1$ distinct real numbers $r_0, r_1, r_2, \dots, r_k$, define the *Lagrange basis polynomial*

$$\ell_j(x) = \frac{(x - r_0)(x - r_1) \cdots (x - r_{j-1})(x - r_{j+1}) \cdots (x - r_k)}{(r_j - r_0)(r_j - r_1) \cdots (r_j - r_{j-1})(r_j - r_{j+1}) \cdots (r_j - r_k)}.$$

The polynomial $\ell_j(x)$ is a polynomial of degree k , as can be seen from the fact that the numerator has k terms, each with an x in it, multiplied together, and the denominator is not zero because the r 's are distinct and the term $(r_j - r_j)$ has been left out.

5. Compute $\ell_j(r_i)$ for any $1 \leq i \leq k$ with $i \neq j$. [1 pt.]
6. Compute $\ell_j(r_j)$. [1 pt.]
7. Given the restriction that Alice must choose a polynomial of degree at most k , prove that Bob can determine her polynomial by asking $k + 1$ questions. [2 pts.]
8. Given the restriction that Alice must choose a polynomial of degree at most k , prove that Bob *cannot* determine her polynomial by asking *fewer* than $k + 1$ questions. [2 pts.]
9. Given the restriction that all the coefficients in Alice's polynomial are nonnegative integers less than 10, Bob asks Alice to compute $p(10)$ and is told that the value is 364,208,592. Determine Alice's polynomial. [2 pts.]
10. Given the restriction that all the coefficients in Alice's polynomial are either 0 or 1, Bob asks Alice to compute $p(2)$ and is told the value is 593. Determine Alice's polynomial. [2 pts.]
11. Given the restriction that all the coefficients in Alice's polynomial are either 0 or 1, explain how Bob can always determine Alice's polynomial by asking Alice to compute the single value $p(2)$. [2 pts.]
12. Given the restriction that all the coefficients in Alice's polynomial are nonnegative integers less than b , explain how Bob can always determine Alice's polynomial with just one question. [3 pts.]
13. Given the restriction that all the coefficients in Alice's polynomial are nonnegative integers, explain how Bob can determine Alice's polynomial with just two questions. [3 pts.]
14. Given the restriction that all the coefficients in Alice's polynomial are either 0, 1, or -1 , Bob asks Alice to compute $p(3)$ and is told the value is 1573. Determine Alice's polynomial. [3 pts.]
15. Given the restriction that all the coefficients in Alice's polynomial are $-1, 0, 1, 2$, or 3 , Bob asks Alice to compute $p(5)$ and is told the value is 1238. Determine Alice's polynomial. [3 pts.]

16. Given the restriction that all the coefficients in Alice's polynomial are integers in the interval (m, n) , explain how Bob can determine Alice's polynomial with just one question. Note: m and n need not themselves be integers. [4 pts.]
17. Problem 13 asked you to find a way to determine Alice's polynomial with just two questions, given that all of her coefficients were nonnegative integers. Now assume that all of her coefficients are integers, the leading coefficient is positive, and the smallest coefficient is known to be at least $-n$ for some positive integer n . Explain how to determine Alice's polynomial with just two questions. [4 pts.]